

Foundation (Jalapeno)

- Q1.** Factor the expression $x^2 - 16$
(A) $(x + 4)(x - 4)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $x^2 - 4x + 4$
(E) $x^2 + 4x - 4$
- Q2.** A basketball player shoots x free throws and makes y of them. If the player misses $x^2 - y^2$ shots, how many shots did the player attempt if $x = 25$ and $y = 20$?
(A) 20
(B) 25
(C) 45
(D) 25 - 20
(E) 625 - 400
- Q3.** Factor $49x^2 - 9$
(A) $(7x - 3)(7x + 3)$
(B) $(7x + 3)(7x - 3)$
(C) $49(x^2 - 1)$
(D) $49(x^2 + 1)$
(E) $9(7x^2 - 1)$
- Q4.** The area of a football field is $a^2 - b^2$ square meters. If $a = 50$ and $b = 30$, what is the area of the field?
(A) 2000
(B) 2500
(C) 1600
(D) 400
(E) 250
- Q5.** A runner completes x kilometers in y hours. The difference of their speeds in km/h is given by $x^2 - y^2$. If $x = 15$ and $y = 10$, what is the difference in their speeds?
- Q6.** Factor $4x^2 - 9$
(A) $2(2x^2 - 1)$
(B) $(2x + 3)(2x - 3)$
(C) $4(x^2 + 1)$
(D) $9(4x^2 - 1)$
(E) $4(x^2 - 1)$
- Q7.** The distance between two basketball players is $x^2 - y^2$ meters. If $x = 8$ and $y = 5$, what is the distance between the players?
(A) 39
(B) 45
(C) 51
(D) 39
(E) 51
- Q8.** A sports team scores x points in the first half and y points in the second half of a game. If the difference of their scores is $x^2 - y^2$, and $x = 20$, $y = 12$, what is the difference of their scores?
(A) 64
(B) 28
(C) 52
(D) 56
(E) 28

Open Ended Questions (FRQ)

- Q9.** A cyclist rides x kilometers in the morning and y kilometers in the evening. If the difference of the squares of the distances traveled is $x^2 - y^2$, and $x = 18$, $y = 10$, find the difference of the squares of the distances.
- Q10.** The time taken to cover x kilometers is t_1 hours and the time taken to cover y kilometers is t_2 hours. If $x^2 - y^2 = 36$ and $x = 8$, find the value of y . Show all steps.

Chef's Tips

- Focus on factoring difference of squares
- Encourage students to identify $a^2 - b^2$ patterns
- Remind students that this pattern applies to various contexts, including sports